

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

E-Commerce Website with Customer Insights and Sales Monitoring Dashboard for R&G Trading.

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rapid development of internet technology has greatly changed how businesses operate and interact with customers. One of the most significant changes is the growth of electronic commerce (e-commerce), which enables businesses to sell products and services online and reach a broader market. E-commerce platforms provide convenience for both businesses and consumers by allowing product browsing, ordering, and communication through digital systems.

However, many small and medium-sized businesses still rely on traditional sales methods and manual record keeping. These practices often make it difficult to monitor transactions, analyze customer preferences, and accurately track sales performance. Without a digital system, businesses may struggle to compete in the rapidly growing digital marketplace.

R&G Trading, like many local businesses, can benefit from adopting an e-commerce system that integrates online selling with customer data analysis. A web-based e-commerce platform with customer insights and a sales monitoring dashboard can help the business manage transactions, analyze customer purchasing patterns, and monitor business performance more effectively.

Therefore, this study proposes the development of an E-Commerce Website with Customer Insights and Sales Monitoring Dashboard for R&G Trading. The system aims to improve the business's ability to manage sales, understand customer behavior, and support data-driven decision-making to enhance overall business performance.

1.2 Statement of the Problem

This study aims to address the following problems:

1. How did R&G Trading improve its current sales process using an e-commerce platform?

2.How did the business effectively track and monitor sales transactions?

3.How were customer purchasing behavior and preferences analyzed efficiently?

4.How did a sales monitoring dashboard support better decision-making?

5.How did the system improve overall business performance and customer satisfaction?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study was to design and develop an E-Commerce Website with Customer Insights and Sales Monitoring Dashboard for R&G Trading to improve online sales transactions, monitor business performance, and support data-driven decision-making.

1.3.2 Specific Objectives

Specifically, the study aims to:

- 1.Develop a web-based platform for online product selling.
- 2.Create a system that records and manages sales transactions.
- 3.Analyze customer purchasing behavior and preferences.
4. Design a dashboard for monitoring sales performance.
5. Improve business decision-making through data analytics.

1.4 Scope and Delimitation

This study focused on the development of a web-based e-commerce website with customer insights and a sales monitoring dashboard for R&G Trading. The system allowed the business to sell products online, manage customer information, and record sales transactions efficiently. It included features such as product listing, online ordering, customer management, and a dashboard that

provided reports on sales performance and customer purchasing behavior. The system analyzed customer data, including purchase history and preferences, to support better decision-making.

However, the study was limited to the operations of R&G Trading only and did not cover other businesses. The system was accessible through web browsers and did not include a mobile application version. Payment methods were limited to basic options such as Cash on Delivery (COD) and simple payment integration. Advanced enterprise-level features were beyond the scope of the study.

1.5 Significance of the Study

The proposed system will benefit the following:

- R&G Trading:** — Improved sales management and business decision-making.
monitoring.
- Customers** — Provided a more convenient shopping experience.
- Researchers** — Served as a reference for future studies.
- Developers** — Offering insights into building e-commerce systems with analytics.

1.6 Definition of Terms

- **E-Commerce**—The process of buying and selling products or services over the internet.
- Customer Insights**—The analysis and interpretation of customer data to understand purchasing behavior, preferences, and needs.
- Sales Monitoring Dashboard**—A visual tool that displays sales data, performance metrics, and business reports in an organized and easy-to-understand format.

•**Web-Based System**—An application that is accessed through a web browser using the internet.

•**Customer Purchasing Behavior**—The patterns and actions of customers when selecting and buying products or services.

•**Data Analytics**—The process of examining data to extract useful information for decision-making.

•**Transaction** – A completed interaction between a customer and the business, such as a product purchase.

•**Product Catalog** – A digital list of products available for sale, including descriptions, prices, and images.

•**User Interface (UI)** – The visual elements of a system that users interact with, such as buttons, menus, and forms.

•**Database** – A structured collection of data stored electronically and used for managing information.

•**Administrator** – A user who manages and controls the system, including products, users, and reports.

•**Cash on Delivery (COD)** – A payment method where customers pay for the product upon receiving it.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

In this study instance, digital analytics tools enable businesses to gain deeper insights into customer behavior by capturing user interactions such as product views, search activities, and purchase transactions.

These behavioral data allow organizations to develop personalized

marketing strategies, enhance customer satisfaction, and strengthen their competitive position in the digital marketplace (Chaffey, 2022). Moreover, integrated e-commerce information systems support essential operational components including user account management, product catalog administration, order processing, payment management, and database-driven reporting, thereby ensuring efficient and reliable business processes (Laudon & Traver, 2021).

Studies focusing on analytics applications in e-commerce emphasize that customer and transactional data are valuable organizational assets. Businesses utilize analytical tools to identify market trends, forecast customer demands, optimize inventory levels, and improve sales strategies. Furthermore, dashboards and reporting systems facilitate real-time monitoring of orders, revenues, and product performance, enabling faster, more accurate, and evidence-based managerial decisions (Turban et al., 2020).

2.2 Local Studies

Several local studies have examined the adoption of e-commerce systems among Philippine businesses, highlighting improvements in sales performance, customer engagement, and data management practices. Research shows that transitioning from traditional manual sales processes to digital commerce platforms enhances operational efficiency and enables more accurate monitoring of customer transactions and purchasing behavior.

Santos (2023) investigated online consumer behavior in Metro Manila and found that the adoption of e-commerce platforms significantly increased sales revenue, strengthened customer engagement, and improved the accuracy of tracking buying patterns. The study emphasized the importance of maintaining well-structured customer and transaction databases to support efficient reporting and informed business decisions.

2.3 Related Literature

Data analytics and e-commerce systems play a significant role in improving business efficiency, decision-making, and operational performance. Studies show that integrating analytics into digital commerce platforms enables organizations to transform large volumes of customer and transactional data into meaningful insights that support strategic planning and performance optimization.

Predictive analytics and supervised learning techniques are widely used to analyze historical transaction data and forecast market trends. These analytical approaches help businesses optimize inventory levels, improve product recommendations, and enhance sales strategies through data-driven forecasting models (Alpaydin, 2020). Furthermore, structured software engineering methodologies are critical in ensuring system quality, maintainability, and reliability. Well-designed systems follow clear architectures that separate user interfaces, server-side processing, and database management (Pressman, 2019).

Moreover, e-commerce platforms that integrate analytics capabilities provide actionable insights that strengthen business strategies. Features such as customer activity tracking, product performance monitoring, order management systems, and sales dashboards enable organizations to adopt data-driven approaches that increase revenue and improve customer satisfaction (Chaffey, 2022; Laudon & Traver, 2021).

2.4 Synthesis of the Reviewed Literature

The reviewed literature consistently indicates that web-based e-commerce systems integrated with data analytics and dashboard monitoring yield measurable improvements in business operations, customer engagement, and decision-making efficiency. Chaffey (2022) emphasizes that platforms with digital analytics tools allow businesses to track customer behavior, optimize product recommendations, and enhance marketing strategies. Similarly, Laudon and Traver (2021) highlight that centralized transaction and product management

systems improve operational efficiency, reporting accuracy, and overall business performance.

Studies focusing on dashboards and reporting tools demonstrate their importance in transforming large volumes of transactional data into actionable insights. Few (2019) notes that well-designed dashboards improve usability, provide real-time reporting, and support faster, evidence-based managerial decisions. Turban et al. (2020) similarly show that real-time monitoring of sales, orders, and product performance allows businesses to respond promptly to market trends and operational needs.

Despite these benefits, the literature reveals limited studies on small and medium enterprises implementing fully integrated e-commerce platforms that combine user management, product catalogs, order processing, customer activity tracking, and analytics dashboards within a single unified system. This study addresses this gap by developing a web-based e-commerce system for R&G Trading that integrates these functionalities into a centralized platform, ensuring streamlined transaction processing, accurate data tracking, and enhanced operational and managerial efficiency.

By combining transaction management, inventory monitoring, customer behavior analysis, and dashboard visualization, the proposed system aligns with current technological standards and modern business practices.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a developmental research design focusing on the design, development, and evaluation of the proposed e-commerce system for R&G Trading Air Conditioner Sales.

3.2 System Development Methodology

The Agile Development Model is adopted in developing the system. The phases include:

- **Requirements Analysis** – Data are gathered through interviews and observation of R&G Trading's current sales process to identify system requirements and existing problems.
- **System Design** – The system architecture, database structure, user interface, and dashboard components are designed based on the identified requirements.
- **Development** – The e-commerce platform is developed, including modules for product management, online ordering, sales tracking, and customer analytics.
- **Testing** –The system undergoes functionality and usability testing to ensure accuracy, reliability, and user-friendliness.
- **Implementation** – The system is deployed for actual use by selected users of R&G Trading.
- Evaluation** – The system is evaluated based on usability, efficiency, and user satisfaction through surveys and feedback, addressing the identified problems in the study

3.3 System Architecture

The E-Commerce Website with Customer Insights and Sales Monitoring Dashboard for R&G Trading follows a three-tier architecture.

Users such as customers, administrators, and store personnel interact with the system through the client-side layer using a web-based interface. The server-side layer is responsible for handling business logic, order processing, user authentication, sales analytics, and dashboard visualization, while the database layer manages and stores all system data including user accounts, products, categories, orders, payments, addresses, and customer activity logs.

This architecture ensures organized system structure, efficient processing of transactions, centralized data management, scalable performance, and secure access for multiple users at the same time.

3.4 Tools and Technologies Used

- **Frontend:** PHP, CSS, JavaScript
- **Backend:** Python / PHP / Node.js
- **Database:** MySQL
- **Development Tools:** Visual Studio Code

3.5 Data Gathering Techniques

- **Surveys and Questionnaires** – To determine customer needs, buying behavior, and usability preferences.
- **Interviews** – Conducted with R&G Trading staff and management.
- **Document Review** – Analysis of current sales reports, transaction records, and existing manual workflows.

3.6 Testing Procedures

- **Unit Testing** – Testing of individual modules such as authentication, product management, and order processing.
- **System Testing** – End-to-end testing for integration, reliability, and performance.
- **User Acceptance Testing (UAT)** – Validation with intended users to ensure requirements are met.

3.7 Evaluation Criteria

The system is evaluated based on:

- Functionality
- Usability
- Security

Data Flow Diagram (DFD – Level 0)

External Entity:

User – Browses products, places orders, and makes payments.

Process:

R&G Trading E-Commerce System—Handles all business operations.

Data Flows:

- Users → browse products → System
- Users → register/login → System
- Users → place orders → System
- Users → make payments → System
- System → update inventory → Database
- System → record transactions → Database
- System ↔ retrieve data → Database

Entity Relationship Diagram (ERD) Entities:

- **Users:** id, email, password_hash, first_name, last_name, phone, role
- **Products:** id, category_id, name, model_number, brand, price, stock_qty
- **Categories:** id, name, slug, description
- **Orders:** id, order_number, user_id, address_id, status, payment_status, total_amount
- **Order_Items:** id, order_id, product_id, quantity, unit_price, total_price
- **Addresses:** id, user_id, street, city, province, zip_code
- **Customer_Activity:** id, user_id, event_type, product_id, created_at

Relationships:

- A user can place many orders.

An order contains many order items.

Order_Items reference Products.

Products belong to categories.

A user can have multiple addresses.

A user generates customer_Activity records.

Use Case Diagram

Actors:

- User
- Administrator

Use Cases:

- Register Account
- Login Account
- Browse Products
- Manage Address
- Place Order
- Make Payment
- View Order History

Administrator Use Cases:

- Manage Users
- Manage Products
- Manage Categories
- Monitor Orders
- View Sales Dashboard
- View Customer Activity Logs

Use Case Diagram

Actors:

- User
- Administrator

Use Cases:

- Register Account
- Create/Edit Profile
- View Matches
- Send Messages
- Monitor User Behavior (Admin)
- Manage Accounts (Admin)

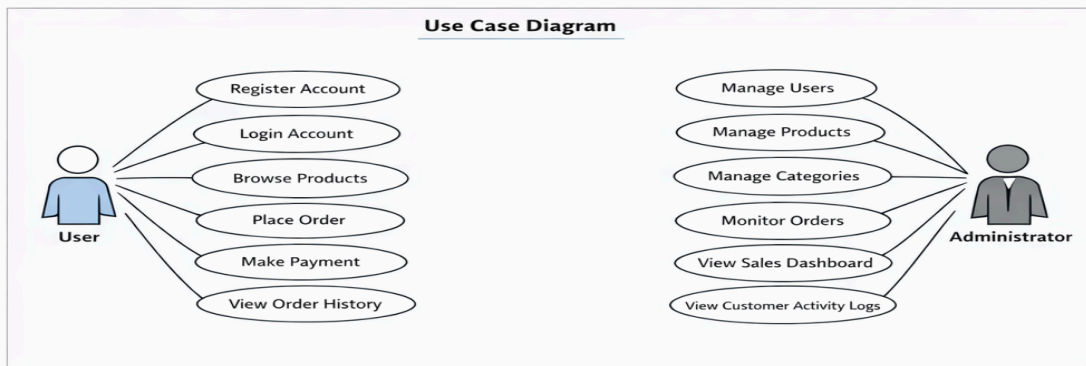
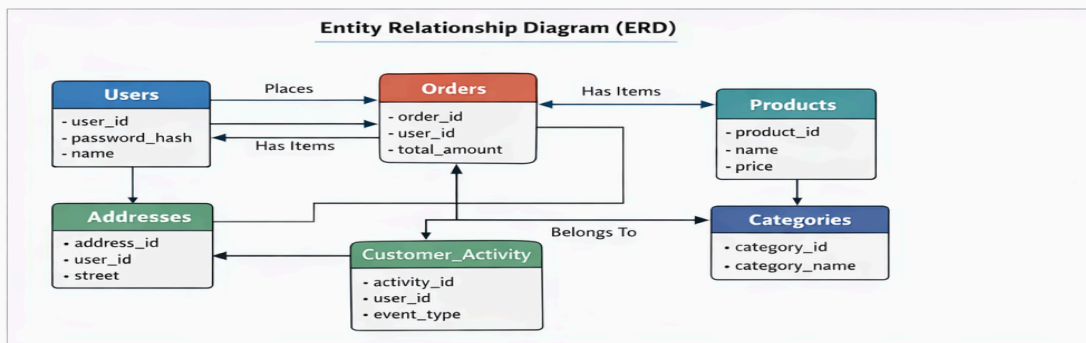
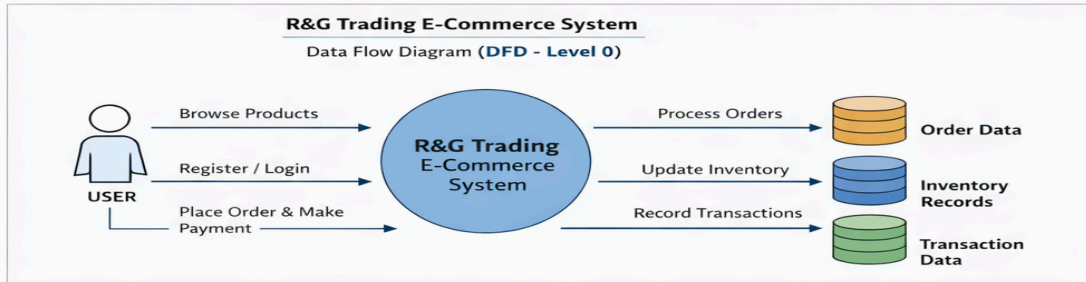
End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.

SUPPORTING DIAGRAMS (Included within Chapter 3)



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